Dean Jie Zhong

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Personal Website↗

Education

BSc Biological Sciences, Xi'an Jiaotong-Liverpool University, China (2023–27, expected)

- Cumulative GPA: 3.74/4.00, Expected First-Class Honours.
- Transcript↗

Research Experience

TroGen: A commensal bacterial platform for intracellular drug delivery in TNBC, iGEM, XJTLU (Feb–Nov 2025)

- Conceptualized and led the project; designed genetic systems and experiments to evaluate their function.
- Established and refined bacteria–cell co-culture workflows, cultured mammalian cells and tumor spheroids, and conducted co-culture experiments. Performed flow cytometry, confocal imaging, and data analysis.
- Wrote scientific wiki content and prepared figures; represented the team in the project presentation and Q&A at the Grand Jamboree in Paris.
- Awards: Top 10 Project (among 190 teams); Best New Composite Part (only winner; undergraduate division).

CD8⁺ T Cell Engineering, Intern, Suzhou Institute of Systems Medicine, CAMS & PUMC (Jan 2026–present)

- Performed CD8⁺ T cell experiments, covering viral packaging; T cell isolation, activation, transduction and expansion; flow cytometry staining, data acquisition and FlowJo analysis; and Transwell migration and LDH cytotoxicity assays.
- Performed animal experiments, including subcutaneous tumor inoculation, tail-vein T cell transfer, ear tagging and animal preparation.
- Performed 293F transfection and protein purification.
- Served as co-first author of a Spotlight article in *Trends in Cancer* (IF 21.6) Spotlight critically analyzing genome-wide CRISPR screening in T cells, linking biological context and functional readouts to guide T cell engineering.
- Proposed, designed and conducted experiments evaluating Cesome as a candidate material for plasmid endotoxin removal.

Motility Mechanisms in Flavobacterium johnsoniae, FYP, XJTLU (Jan 2026–present)

- Conceptualized and led experimental development of a fluorescence-based "founder-equivalent mass" approach to distinguish bacterial migration from cell proliferation; also explored fluorescent labeling of colony polysaccharides.
- Proposed an alternative labeling strategy for SprB and led antibody-labeling trials using strains expressing sfGFP–SprB fusions, assessing labeling specificity and reliability for single-particle tracking.
- Built a Python workflow to automatically generate temporal color projections from microscopy videos; developed and iteratively tested a single-cell tracking pipeline designed to maintain cell identities through interactions and support reliable motion analysis.

Publications

Zhong, J.*, Wang, J.*, Zhang, L. † (2026). Genome-wide human T cell screens: barriers and design. *Trends in Cancer*. In press. * *Equal contribution; † Corresponding author.*

Honors & Awards

TroGen: A commensal bacterial platform for intracellular drug delivery in TNBC, iGEM, XJTLU (Feb–Nov 2025)

- Top 10 Project (out of 190 teams).
- Best New Composite Part (Only Winner; Undergraduate Division).
- Best Oncology Project & Best Wiki (Nominated).
- Gold Medal.

Third Prize, China High School Biology Olympiad (Nov 2022)

Academic Mentorship

Advisor, Zhejiang University–University of Edinburgh Institute iGEM Team (Jul 2026–present)

- Advised on project design, experimental planning, and laboratory techniques; provided strains and plasmids.

Instructor, Soochow University ICPI iGEM Team (May 2026–present)

- Founded the college's first iGEM team and secured university funding; advised on project design, experimental planning, and laboratory techniques; facilitated external collaborations.

Advisor, Xi'an Jiaotong-Liverpool University iGEM Team (Jan 2026–present)

- Supported recruitment, guided project conceptualization, assisted in securing university sponsorship; advised on project development and facilitated external collaborations.

Private Tutor, Self-Employed, Suzhou, China (Jul–Sep 2023)

- Subjects: Biology, Physics, English.

Presentations & Posters

Zhong, J. (2026). *A commensal bacterial platform for intracellular drug delivery in TNBC*, Poster, BIOCHINA Youth Research Exhibition, Suzhou, China (14 Mar 2026)

Zhong, J., et al. (2025). *A commensal bacterial platform for intracellular drug delivery in TNBC*, Presentation, Conference of China iGEM Community & Global Bio-Developer Conference, Beijing, China (7 Aug 2025)

Zhong, J. (2025). *A commensal bacterial platform for intracellular drug delivery in TNBC*, Presentation, School of Science XJTLU, Suzhou, China (20 Oct 2025)

Additional Experience

Visiting Research Student, Qingdao Institute of Bioenergy and Bioprocess Technology, CAS (Jul 2025)

- Led technical discussions on restriction–modification systems, environmental sensing, and drug-release strategies with institute researchers; contributed to establishing an effective genetic transformation method for *Staphylococcus xylosus*.

Member, Suzhou Society for Biology (May 2025–present)

Class Representative & Student Community Representative, XJTLU (Sep 2023–Jun 2024)

Technical Skills

Languages:

- **English:** IELTS Academic 8.0, Transcript↗ (Aug 2026).
- Chinese (Mandarin): Native; Chinese (Hakka): Conversational.
- Spanish: Basic.

Experimental Techniques:

- **Mammalian Cell Culture & Engineering:** Mammalian cell culture; tumor spheroid culture; viral packaging, T cell isolation, activation, viral transduction and expansion; 293F transfection.
- **Microbiology:** Bacterial culture, bacteria–mammalian cell co-culture; bacterial transformation.
- **Animal Procedures:** Mouse dissection, tissue harvesting; ear tagging, subcutaneous tumor inoculation, tail-vein injection.
- **Cellular & Protein Techniques:** Flow cytometry, Transwell migration assay, LDH cytotoxicity assay, affinity chromatography, protein purification, Western blotting, ELISA.
- **Microscopy & Imaging:** Confocal microscopy, TIRF microscopy, time-lapse and z-stack imaging, quantitative image analysis, 3D reconstruction.
- **Molecular Biology:** PCR, gel electrophoresis, Gibson Assembly, Golden Gate Assembly.
- **Biophysical Techniques:** Spectrophotometry, Dynamic Light Scattering (DLS).

Scientific Software: Imaris, Fiji/ImageJ, FlowJo, and PyMOL.

AI-Assisted Software Development: Software design and iterative development using Codex and Claude Code.

Scientific Visualization: BioRender, Adobe Illustrator, Adobe Photoshop.

Knowledge Management: local LLM deployment and retrieval-augmented generation (RAG) systems.

Extracurricular Activities

Sports

- Road Cycling: Road Race 47th, Individual Time Trial 35th, China U&C Student Cycling Challenge (May 2024).
- Basketball: Middle & High School Varsity Teams (2017–23); XJTLU School of Science Team (2023–25).
- Strength Training & Calisthenics.

Music

- Chinese Bamboo Flute (竹笛), Chinese vertical bamboo flute (萧), and Tin whistle.